

Akmal Ashirmatov

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M.Sc. Computer Science, ETH Zürich | LLM post-training, test-time scaling, agentic systems, multimodal generation

EDUCATION

- **ETH Zürich** *M.Sc. Computer Science, GPA 5.93/6.0* *Sep. 2024 – Apr. 2027 exp.*
 - Selected coursework: Probabilistic AI, Advanced ML, Reliable and Trustworthy AI, Large-Scale AI Engineering, Computer Vision, Algorithms Lab.
- **Technical University of Munich** *M.Sc. Data Engineering and Analytics, 1 semester* *Apr. 2024 – Sep. 2024*
- **Lomonosov MSU Tashkent** *B.Sc. Applied Mathematics & CS, with Honors, GPA 5.00/5.00* *2018 – 2022*

PUBLICATIONS

- **On Repulsive and Attractive Teachers: Separating Correctness from Behavior in Self-Distillation** [↗](#) *FLLMPT @ NeurIPS 2026* *2026*
 - Anton Baumann, **Akmal Ashirmatov**, Leo Schmidt-Traub, Frederike Lübeck, Jonas Hübotter, Thomas Kleine Buening, Andreas Krause
 - Studied attractive, repulsive, and contrastive self-distillation, showing that contrastive teachers cancel unintended behavioral shifts and improve reasoning.
- **Test-Time Scaling with Weak Verifiers via Self-Play** [↗](#) *DL4C @ ICML 2026* *2026*
 - Akira Yoshiyama*, **Akmal Ashirmatov***, Jonas Hübotter, Ido Hakimi, Andreas Krause
 - Self-play proposer-solver framework for improving code-generation models with weak verifier and environment feedback.
- **ESCAPE – Explaining Stable Diffusion via Cross Attention Maps and Prompt Editing** [↗](#) *VISxAI 2025* *2025*
 - Diego Zafferani*, Giovanni De Muri*, Johanna Hedlund Lindmar*, **Akmal Ashirmatov***, Sinie van der Ben, Rita Sevastjanova, Mennatallah El-Assady
 - Built an interactive system for cross-attention visualization and Prompt-to-Prompt-style localized image editing.

EXPERIENCE

- **SWE Intern, Applied Research team** *Google, London* *Jun. 2026 – Sep. 2026*
 - Designed and prototyped a new generative tool for Gemini Live and contributed code to the Gemini app codebase.
 - Authored a design document for integrating the tool into the existing agent harness.
- **Research Intern, Zurich Research Center** *Huawei* *Jun. 2025 – Sep. 2025*
 - Worked on LLM inference optimization using Triton and vLLM; performed hardware profiling, ML compiler optimization, and IR-level debugging.

SELECTED PROJECTS

- **Distributed PEFT for Legal LLMs** [↗](#) *Apertus 8B/70B* *2026*
 - Built a PEFT/SFT pipeline with LoRA and full fine-tuning; ran multi-node Accelerate + DeepSpeed training with DDP/ZeRO-2/ZeRO-3.
- **Network Certification & Robustness** *Reliable and Trustworthy AI* *2026*
 - Certified encoder/decoder properties via MILP (Gurobi) and DeepPoly relaxations; added PGD-based adversarial defense training.
- **3D Vehicle Detection from LiDAR Point Clouds** *1st place in class* *2026*
 - Redesigned second-stage RoI refinement with proposal-aligned canonical transforms, point-wise/geometric features, and IoU-aware scoring.
- **Music Semantic Search** *2025*
 - Trained retrieval embeddings from CQT spectrograms using triplet loss with hard negative mining; an end-to-end cover-song retrieval pipeline.
- **Text Classification with Test-Time Augmentation & Selective Reasoning** [↗](#) *2025*
 - Applied TTA via 4 paraphrases with weighted ensembling. Added selective LLM reasoning for uncertain cases

SKILLS

- **ML/LLM:** *PyTorch, Hugging Face, vLLM, verl, GRPO/RLHF-style training, LoRA/PEFT, W&B*
- **Systems:** *Python, C/C++, Triton, Bash, Linux, Docker, Kubernetes, Ray, DeepSpeed, Spark, GCP, HPC/Slurm*

AWARDS & COMPETITIONS

- ICPC Uzbekistan: 1st, 2nd, 1st, 1st place; 3-time National champion *2019 – 2022*
- ICPC NERC: 2× 2nd-degree diploma; 4th and 5th in Central Asia *2021 – 2023*
- Codeforces: Master, peak rating 2124 [↗](#)
- Meta Hacker Cup top 1.3%, Google Code Jam top 1.9%, CodeChef SnackDown top 0.8% *2021 – 2022*
- Huawei iRunner: 6× prize-winner across 8 countries; high-performance heuristic optimization *2022 – 2025*